PREVIOUS YEARS QUESTIONS

I. SHORT ANSWER TYPE QUESTIONS

Q.1 ***Define steam boiler and write different types of boiler.*** *[R.T.U. 2018]*

Ans. A steam generator or a steam boiler is a steel enclosure used to transfer the heat generated by the burning of fuel (solid, liquid, gas) to water, which leads to production of steam.

Types of Boiler : Boilers are generally classified as follows :

(a) Fire tube and water tube boilers

(b) Horizontal, vertical and inclined

(c) Externally fired and internally fired

(d) Locomotive, marine, stationary and portable

(e) Natural circulation and forced circulation

Q.2 ***Write down the Differentiate between water tube boiler and fire tube boiler.*** *[R.T.U. 2018]*

Ans. Water Tube Boiler

1. Water is circulated inside the tubes surrounded by hot furnace gases.
2. Steam pressure is higher and goes upto 165 bar.
3. Rate of steam generation is as high as 450 tonnes per hour.
4. Overall efficiency reaches up to 90% with economizer.
5. Lesser area is required to generate a given amount of power, around 5 meter square per tonne per hour.
6. It is separable into its various components and hence can be transported and assembled easily.
7. It can be operated easily at fluctuating load.
8. Water circulation is done in a well-defined direction.
9. There are chances of bursting.
10. Bursting does not lead to complete destruction of the boiler.
11. The operating cost is high.
12. It is suitable to use in large power plants.

Fire Tube Boiler

1. Hot gases from the grate are passed through the tubes surrounded by water.
2. Steam pressure can rise only up to 24.5 bar.
3. About 9 tonnes of steam per hour are generated.
4. Overall efficiency reaches up to 75%.
5. To generate given power, more floor area (about meter square per tonne per hour) is required.
6. It is difficult to transport and set up.
7. It can handle fluctuations only for a shorter perio
8. Direction of circulation of water is not fixed.
9. There is lesser probability of bursting.
10. If bursting occurs, the damage to the property vast.
11. Operating cost is low.
12. It cannot be used for large plants.

Q.3 ***Write down the various factors affecting the selecti of steam boiler.***

Ans. Selection of Steam Boiler : The following fact affect the selection of a steam boiler:

1. Availability of fuel and water.
2. Type of fuel used.
3. Expected load factor.

4. Suitable geographical location for power house.
5. Amount of power and working pressure required.
6. Rate of steam generation.

Q.4 ***Name the different parts of a Cochran boiler.***

Ans. Parts of a Cochran Boiler

(1) Cylindrical shell
(2) Fire box
(3) Flue pipe
(4) Combustion chamber
(5) Smoke tubes
(6) Chimney
(7) Manhole
(8) Grate

II. Long Answer Type Questions

Q.1 ***What are the essential requirements of a good steam boiler?***

Ans. Essential Requirements of a Good Steam Boiler

1. The installation costs should be economical.
2. It should be light in weight and occupy less space.
3. It should start quickly.
4. It should be able to adjust quickly to the fluctuations in load.
5. It should be fuel efficient and produce maximum amount of steam.
6. Heating plates should not be covered by mud and waste deposits.
7. Tubes should not be choked by soot or other deposits. They should have enough strength to withstand considerable amount of wear and corrosion.
8. Refractory material produced should be just in the right amount to sustain easy ignition and prevent smoke from burning the fuel.
9. The amount of maintenance required should be minimum.
10. There should be lesser and easily accessible joints.
11. Frictional losses due to movement of water and flue gas in the circuits should be minimal along with optimum fluid velocity.
12. It should follow all the safety rules as per the Boilers Act.

Q.2 ***Discuss the important terminology related to steam boilers.***

Ans. Important Terminology of Steam Boilers : From the subject perspective, it is important for one to know and understand the important terminology related to the steam boiler. The essential terms related to steam boilers are listed below :

1. **Boiler Shell :** This cylindrical shell is formed by welding or riveting the steel plates whose ends are sealed by end plates. It is used to store the water to be heated. Also, it contains the steam that is formed as a result of heating. Therefore, it should have sufficient storage capacity.
2. **Combustion Chamber :** The combustion chamber, usually located below the boiler shell, is used to burn fuel so that water in the boiler shell can be heated to produce the steam.
3. **Grate :** A grate acts as a platform inside the combustion chamber to burn the coal or wood fuel. It is made from cast iron bars, the space between which allows the air to pass for combustion. The total surface area of the grate on which combustion takes place is called grate surface.
4. **Furnace :** It is located between the grate and the boiler shell. The fuel is actually burnt here. It is also known as fire box.
5. **Heating Surface :** The portion of the boiler surface exposed to fire and hot gases is called heating surface.
6. **Mountings :** Certain components are fitted on the boiler to ensure proper functioning and safety. These are called mountings. For example- safety valve, water level indicator, pressure gauge etc.
7. **Accessories :** These are the additional devices used with the boiler without mounting them on it. These are used to increase the efficiency of the boiler. For example- superheater, economizer, feed pump etc.

Q.3 ***Explain Lancashire boiler with neat sketch.***

Ans. Lancashire Boiler : It is a horizontal axis, stationary fire tube boiler that uses natural circulation of water and flue gases. Heat is transferred through convection and a low pressure steam is generated. This steam is used to drive steam turbines which require moderate power.

The boiler has an external shell in the shape of a cylinder of 1.75-2.75 m in diameter and 7.25-9 m in length. It is made of steel plates which are built in sections and then joined by riveting. Inside the shell there are two flue tubes with diameter about 40% of the shell diameter. The diameter is lesser at the back end to easily reach at the lower part of the boiler. Fuel is burnt in the fire grate located inside the boiler at the end of the flue tubes. A brick arch at the end of fire grate deflects the gases upwards.

The hot flue gases enter into the bottom tube after crossing the internal tubes. These are then passed to the boiler from where they divide into the side flues. The gases finally enter into the main flue tubes and exit through the chimney.

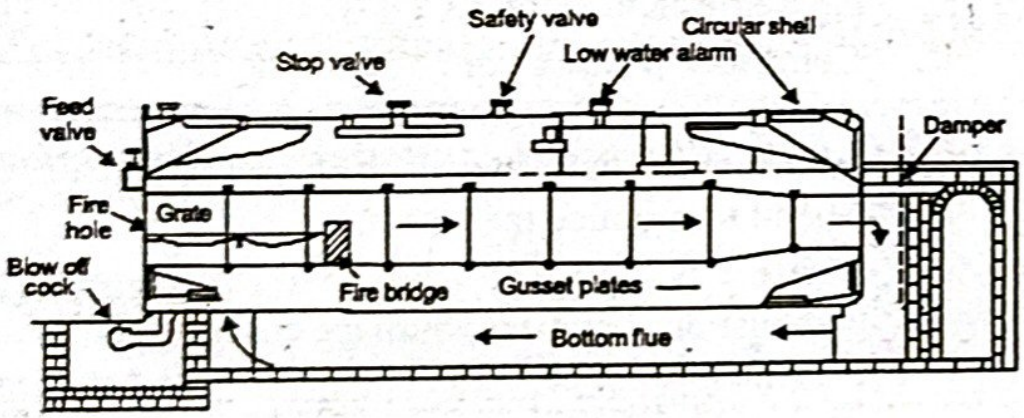

Fig. : Lancashire Boiler

Structures called dampers are operated through a chain-pulley system in front of the boiler. These dampers regulate the amount of steam generated by controlling the rate of air flow (draught) from the ends of side flues.

A spring loaded safety valve is installed on the external shell. Another valve is mounted to ensure high steam and low water. Steam is fed to the engine through the stop valve. Water is fed uniformly through a perforated feed pipe.

Upon excessive heating, the steam space is filled with large quantities of water along with steam. This process is called priming. To separate water, an anti-priming pipe is used and thus stop valve receives dry steam.

A blow-off cork is positioned beneath the front face of the shell. It is used to remove mud and sediments. It is also used to drain out water for inspection. Manholes are present at the top and bottom of the boiler to facilitate cleaning and repairing.

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